

Erik Hill

AI Evaluation & Testing Engineer

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SUMMARY

Self-taught AI evaluation and testing engineer building deterministic, replayable systems for measuring coding agents and LLM applications.

I build **verifiable-evaluation systems** whose claims are checked rather than asserted: **eleven published evaluation claims are replayed from pinned artifacts in CI**, deterministic, no LLM judges, failures published by design. Also **shipped a game to Google Play; 30+ public repos, two PyPI packages, one npm package**.

SKILLS

Agentic / AI: Multi-agent orchestration · LangGraph · Model Context Protocol (MCP) · LLM application development · Tool-calling & guardrails · RAG · Evals · Regression gating (nDCG) · Prompt engineering · Context engineering · Safety (OWASP LLM Top 10, human-in-the-loop) · Differential-oracle testing · Property-based testing · Mutation testing · No-LLM-judge grading · Pre-registration & minimum-detectable-effect analysis

Backend & data: Python · FastAPI · Pydantic · SQLAlchemy 2.0 · PostgreSQL · pgvector · Alembic · REST / OpenAPI

Frontend / mobile: TypeScript · JavaScript · React · Next.js 16 · Java · Capacitor · Android (adb) · WebAssembly (Pyodide) · TeaVM (JVM to JavaScript)

Ship & verify: pytest · Vitest · JUnit · jqwik · Playwright · PyPI / npm packaging · GitHub Actions (matrix CI) · Docker · Cloudflare Pages (Wrangler) · Neon · Test automation · QA for LLM systems

SELECTED PROJECTS, all public at github.com/egnar09

- **agent-certlab:** Two real coding agents certified, every verdict regradeable from artifacts. A certification lab built on three **seeded-defect task families** with provable answer keys (incl. **coordinated two-file defects** no single-file patch can pass), a **precondition prover** (clean passes, every seed fails, it refused one of my own defects), and **calibration agents** the harness must separate before certifying.
- **reference-fleet:** An answer key for eval suites: six models each broken **one documented way at an exactly reproducible seeded rate**, so detection rates are measurable against ground truth, a naive suite detects **1 of 6** (paired protocol). Plus a **LoRA-trained native defect**: mixture 0.507 → realized **0.20 greedy, 0.31 at temp 1.0**: training's rate control is lossy, measured.
- **vac-protocol:** **Verifiable Agent Claims**: an evidence-bundle format + **offline verifier**; a registry where **eleven real claims are hash-checked and re-earned in CI**, with a **public record of it rejecting my own evidence**. Plus **vac-gate**: replayable evidence or no green check.
- **evalmut:** **Mutation testing for eval graders (PyPI, with its gradecore engine)**: inject known defects, report which checks stayed green. Found **4 holes in its own engine's suite, 8 in ports of a widely-used framework's assertions. No LLM judge**; every operator tied to a documented real failure. *319 tests*.
- **rag-eval-lab:** deterministic RAG **eval harness** that catches a planted hallucination in CI; from-scratch **BM25 matching the published SciFact baseline (0.664 vs 0.665)**. *45 tests*.
- **agent-graph:** a **LangGraph ReAct agent** whose multi-step tool use is **unit-testable**: a swappable mock/LLM policy so the graph runs deterministically without a key, a **safe AST calculator** instead of eval, and a max-step guard. *31 tests*.
- **model-drift:** public board, **frozen hash-fingerprinted suite across 17 models / 5 labs**, with a derived **minimum-detectable-regression floor**.
- **harness-builder:** **More scaffolding scored worse, for 22x the cost**. One sweep of one 20-task suite across three harness shapes on real models, \$0.99, **zero errors in 60 runs**: one drafter 95%, planner plus drafter 90%, planner plus two drafters plus judge 80%. The four-call panel beat the single drafter on **0 of 20 tasks at 22.1x the cost and 8.4x the latency**, and the tool reported **"this suite cannot decide"** rather than ranking, because 17 ties leave three informative tasks.
- **tapdodge-engine:** **One rulebook, two runtimes, and a check that fails when they stop agreeing**. The cross-runtime differential is wired into CI, so a disagreement between the JVM engine and the TeaVM-compiled browser build fails the build. In a committed fault injection, the differential was the **only** check that caught a runtime-supplied-RNG fault: the same-runtime determinism test and the golden file both passed it.
- **Also public:** **crashkit** (live) · **mcp-tools** (MCP server, no SDK) · **evals-differential-oracle** (two independently written implementations agree across **3,000 seeded boards**) · **llm-wire-stub** (npm) · a **merged PR to Promptfoo**.

REVIEWED BY OTHERS

- **External audit of vac-protocol:** an outside engineer falsified my own verifier, cheapest forgery **four bytes**. Fix merged, then I found the same defect class **four more times** (arXiv, below). **Sole maintainer through the audit:** merged four of his pull requests and answered his four specification issues in SPEC.md (protocol v0.2). On two I showed the fix he proposed would not close the hole and shipped a wider one, reasoning public.
- **Peer review with Bartłomiej Nawara, PhD**, senior AI/ML engineer: his missing-not-at-random critique of my rate-limit handling shaped **model-drift's reliability floor**, credited in the repo. I found **two design faults** in his pre-registered study, both adopted.
- **Closed a bug report open in TeaVM since 2018** that its own suite could not catch: a **one-character error** in TeaVM's GregorianCalendar emulation, found by differential-testing the browser port against the JVM. The four assertions covering that line had been commented out; the patch re-enables them. Merged same day by the maintainer.

PUBLICATIONS

“Four Ways to Forge a Bundle My Own Verifier Calls Clean: Refusal-Site Mutation Testing of an Evidence-Bundle Verifier.” Erik Hill. **arXiv preprint, 2026**. [arXiv:2608.26183](https://arxiv.org/abs/2608.26183)
21 articles on evaluation and testing failures, **88 comments from 26 readers**. dev.to/agentdev9 · **Drift Notes:** erikhill.dev/notes

EXPERIENCE

Independent AI Evaluation & Testing Engineer · Solo Practice · Remote · Mar 2026 – Present

- Designed **five roles across a four-stage production loop** (Strategy → Execution → Critic → Ops) that plans, builds, reviews, and validates changes to a real React / Capacitor product, human-in-the-loop on every consequential step, plus an **asynchronous Evaluation harness** that scores completed batches.
- Built per-role model routing and hook-based automation that hands work between stages with no manual relaunch; enforced rules with **property-based invariants** (jqwik) over a native engine and **adb-driven on-device validation**. Write-up: github.com/egnaro9/agent-ic-dev-harness
- **Operate model-drift as a running service, not a demo:** its scheduled probe has fired on every calendar day since 2026-07-19 (72 consecutive days as of 2026-09-28, 71 with a clean run), publishes a public board, and files its own regression tickets, **69 drift issues** opened by the pipeline.
- Diagnosed an apparent **66-point overnight drop** on that board (69% to 3% on a frozen suite) as a **provider rate cap rather than model drift**: 34 of 35 calls hit a 429 against a 30 RPM shared tier. Shipped Retry-After-aware retries, a per-host throttle, and a reliability floor that drops low-reliability probes from the accuracy line, so a rate limit can no longer render as a regression.

Indie Game Developer · SeraphLight Studios · Remote · Nov 2025 – Present

- **Shipped Tap Dodge Rush to Google Play** (Kotlin / Java, Android Views + Canvas, AdMob), closed testing with 12+ testers, then store review, then launch. Building a second, larger game (React / Capacitor), the product the harness above operates on.

Kitchen Manager / Shift Lead · Big Island Brewhaus & other high-volume kitchens · Hawaii · 2018 – 2025

- Led shifts and trained staff; owned prep, inventory, and food-safety under sustained pressure, the reliability I bring to software now.

EDUCATION

Clark College, 2007–2009 (some college, no degree) · **freeCodeCamp**: self-directed software curriculum.